

HANDOUTS

115.109

Introduction to Business Analytics

Topic 6: Loyalty; Probability, Proportions & Chi-square

Dr Andrew Murphy

(Please contact me via Communication Tools in Stream)

Week (starting)	Business Problem	Data & Analysis	Chapter (Black et al)	Assessments
1 (13/7)	Understanding Data: discounts & margin	Calculating data in Excel; GST, discounts and Margins	1	Lab1
2 (20/7)	Operational Monitoring: Dashboards	Tabulating & Graphing	2	Lab2
3 (27/7)	Industry trends	Statistics NZ: retail survey Descriptive stats: averages	3	Lab3
4 (3/8)	Customer trends	Statistics NZ: census Descriptive stats: variation	3	Lab4
5 (10/8)	Transaction data: product sales	Transaction data sources. Sampling	7	Lab5
6 (17/8)	Loyalty cards	Loyalty card schemes. Probability; proportions, χ^2	4, 12	Lab6 Test 1

Mid	Semester	Study	Period	
7 (7/9)	Segmentation	Statistics NZ: census. Normal distribution; Means	6, 8	Lab7
8 (14/9)	Determining difference (1)	Means; ANOVA	9, 10, 11	Lab8
9 (21/9)	Associating and Predicting	Explaining relationships: Correlation & regression	13	Lab9
10 (28/9)	Determining Difference (2)	Explaining relationships (continued)	14	-
11 (5/10)	Trends over time	Time series, Indices Unstructured data: social media	15	Lab11 Test 2
12 (12/10)	Corporate reporting	Finance Dashboards reporting		Report

2026	Assessment	Lab 1&2	Lab 3&4	Lab 5&6	Test 1	Lab 7&8	Lab 9,11	Test 2	Report
Week	13-Jul	O: 13.07							
commencing	20-Jul		O: 20.07						
Monday:	27-Jul								
	03-Aug	D: 03.08							
	10-Aug			O: 10.08					
	17-Aug		D: 17.08		O: 14.08 C: 21.08				O 17.08
	<i>Mid-Semester Study Period</i>			D:31.08					
	07-Sep					O: 07.09			
	14-Sep								
	21-Sep						O: 21.09		
	28-Sep								
	05-Oct					D: 05.10		O: 02.10 C: 09.10	
	12-Oct						D: 19.10		D: 19.10
O=opens (available from); D=due (without late penalty); C=closes (not available after)									

Labs 1&2; 3&4; Test 1

- Labs 1&2 mean mark 8.6 (from N=157) (2502: 9.1).
♂ $\bar{x}$: 8.8; ♀ $\bar{x}$: 8.4
- Labs 3&4 file submission was due 17 August (N=125); late penalties now apply
- Model answers to be available next week
- Lab3RBQ and Lab4RBQ can still be completed (N=113,105)
- Test 1 currently underway (N=47): closes Friday 21 August 5pm
- Labs 5&6 underway (due 31 August)

Group Information Survey

☰ Contents

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- ↓ [Assessment Planner](#)
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- ↓ [Missing AI Declaration](#)
- ↓ [Labs 1 & 2](#)
- ↓ [Labs 3 & 4](#)
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- ↓ [Optional Reviews](#)
- ↓ [Assessment 4 information](#)
- ↓ [Analytical Report](#)
- ↓ [Report Groups](#)

56 groups
so far

31 since 10/8



QUESTIONNAIRE

Group Information Survey (for Distance students): closes 21 August (late responses)

If you have not already formed a group or responded to the previous survey, please complete this survey by **Friday August 21**

This survey will help me generate teams of similar distance learners for Assignment 4 (group report). All information in the survey is confidential and is used solely for the purpose of organising teams for this assessment.

Teams created using the survey information tend to be better organised than teams randomly assigned.

Report Groups

- Report details now released
- Report groups determined based on responses to the Group Information Survey or randomly
- Please contact your fellow group members via the In-Group Chat forum (Keeping In Touch section).
- **By 7 September** please complete [Group member activity \(up to September 7\)](#) so I know whether or not you are in contact with your group mates

 FEEDBACK
Group member activity (up to September 7)

DIALOGUE

 **In-Group Chat**

Please indicate whether you have been in contact with your fellow group members, or would like to have your group reformed.

Use this to initiate contact with your group mate(s). Click Create to start a new chat, search for their name (see [Group List](#)), and select the appropriate student name; you can add more than one student to the chat if you are part of a group of 3.

{Unfortunately almost all student names can be selected here, since you are all part of the Auckland Internal student group, or the Distance student group. Please only initiate a chat with students in your group here.}

Group Report

115.109 Report: 'Sustainable Retailing'

This is a group assessment, due before 11pm on Monday October 19 (beginning of exam study break).

It is worth 20% of the final mark, plus an additional 10% allocated to an individual reflection on the group dynamics.

Learning Outcomes Assessed:

(these are **skills** you should be able to demonstrate in your report)

1. Outline how data can be converted into meaningful information, and patterns and opportunities recognised within that data.
2. Identify how information can be interpreted to inform and improve business decision-making.

Using this data and literature, answer *each* of the following:

1. Assess respondent self-reported 'literacy' levels regarding sustainability: how does literacy vary across subgroups (of your choice from the provided dataset)
2. How does 'sustainability literacy' interact with measures of sustainability attitudes and behaviour?
3. What recommendations do you have for a specific (large) NZ retailer (of your choice) to act on these attitudes/behaviours/literacy?

Report: Sustainability attitudes in retailing

- Due by October 19 (can be submitted anytime prior)
- Group report worth 20% + individual reflection worth 10%
 - 1) Assess respondent self-reported 'literacy' levels regarding sustainability: how does this vary across subgroups (of your choice from the provided dataset);
 - 2) How does 'sustainability literacy' interact with measures of sustainability attitudes and behaviour?
 - 3) What recommendations do you have for a specific (large) NZ retailer (of your choice) to act on these attitudes/behaviours/literacy?
- Mixture of
 - use of a provided dataset (SustainabilityData) to create graphs and tables to illustrate an issue
 - short literature review (of academic journals and trade information) to explore the issue
- **Maximum 1500 words across all sections** (excluding Executive Summary, References and Appendices)
- One report submission for the group
- Each group member to separately submit an individual reflection on how the group functioned (eg how tasks allocated, how any conflicts were managed)
- Report instructions in Assessments section

(We use SustainabilityData in Labs 1 & 2, 4, 5 & 6, 7 & 8)

Consumer Purchase Probabilities

- How often do customers shop at a particular store?
- If they shopped last week, will they shop next week too?
- If they are a frequent shopper, will they shop next week?
- If they are a high-spending shopper, will they shop next week?

RFM stands for the three dimensions:

- ✓ **Recency** – *How recently did the customer purchase?*
- ✓ **Frequency** – *How often do they purchase?*
- ✓ **Monetary Value** – *How much do they spend?*

Fader, P.S., Hardie, B.G., & Lee, K.L. (2005). RFM and CLV: Using iso-value curves for customer base analysis. *Journal of Marketing Research*, 42(4), 415-430

- If they purchased product A, will they also purchase product B?

The Probability of Probability

- In everyday language we often refer to the *probability* that certain events will happen
- We also use the word ‘chance’ as a substitute for probability on some occasions
- While we all use the word ‘probability’ in our language, there would be few people who could provide a formal definition of its meaning
- *Examples:*
 - There is a 10% chance that it will rain tomorrow
 - There is a 2-in-6 chance the All Blacks will lose to France in the Rugby World Cup quarter/semi/finals
 - There is a 50:50 chance that she loves me (“she loves me not”)
 - There is a 0.1% likelihood that a call to Spark will connect with a live person within 1 hour and 10% within 2 hours *

(* felt experience of at least one customer)

The Probability of Events

- *Sample space*
 - When a statistical experiment is conducted, there are a number of possible outcomes
 - These possible outcomes are called a *sample space* and this is denoted by S
 - E.g. a coin is tossed. What is the sample space?
 - Solution: $S = \{\text{head, tail}\}$
- *Events*
 - An *event* is a specified subset of a sample space
 - E.g. a coin is tossed. Define event A as the outcome 'heads'
 - Solution: $A = \text{outcome is a head}$

Example: Tossing Coins

- *Toss a coin and record whether a Head or Tail*
 - First toss outcome
 - Second toss outcome
 - Third toss outcome
- What is the probability of a Head in the first toss?
- What is the probability of a Head in the second toss?
- What is the probability of a Head in either the first or second toss?
- What is the probability of a Head in the second toss, if the first was a Head?

The Probability of Events

- *Probability*

- If A is an event, the *probability* that it will occur is denoted by $\Pr(A)$
- The *probability* (or *chance*) that an event A will occur is the proportion of possible outcomes in the sample space that yield the event A . That is:

$$\Pr(A) = \frac{\text{number of outcomes that yield event } A}{\text{total number of possible outcomes}}$$

- The definition makes sense only if the number of possible outcomes (the sample space) is finite
- If an event can never occur, its probability is 0. An event that always happens has probability 1
- The value of a probability must always lie between 0 and 1
- A probability may be expressed as a decimal or a fraction

Mutually Exclusive Events

- Two events A and B are said to be *mutually exclusive* if they cannot occur simultaneously
- If two events A and B are mutually exclusive, the following relationship holds:

$$\Pr(A \text{ or } B) = \Pr(A) + \Pr(B)$$

- Suppose that $A_1, A_2, A_3, \dots, A_n$ are mutually exclusive events. Then:

$$\begin{aligned} &\Pr(A_1 \text{ or } A_2 \text{ or } A_3 \dots A_n) \\ &= \Pr(A_1) + \Pr(A_2) + \Pr(A_3) + \dots + \Pr(A_n) \end{aligned}$$

Independent Events

- Two events A and B are *independent events* if the occurrence of one does not alter the likelihood of the other event occurring
- Events that are *not* independent are called *dependent events*
- If two events A and B are independent, the following relationship holds:

$$\Pr(A \text{ and } B) = \Pr(A) \times \Pr(B)$$

- Suppose that $A_1, A_2, A_3, \dots, A_n$ are n independent events

- Then
$$\begin{aligned} &\Pr(A_1 \text{ and } A_2 \text{ and } A_3 \dots \text{and } A_n) \\ &= \Pr(A_1) \times \Pr(A_2) \times \Pr(A_3) \times \dots \times \Pr(A_n) \end{aligned}$$

continued

Complementary Events

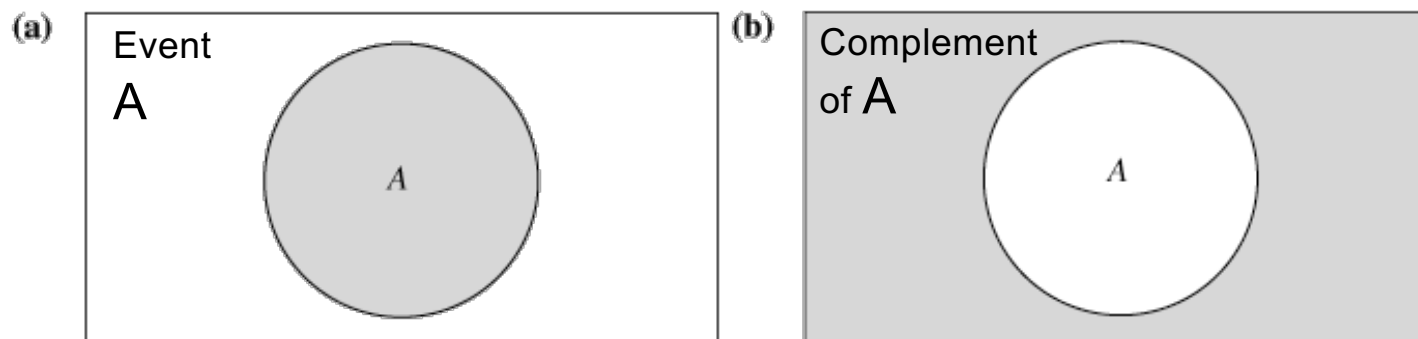
- Complementary events are mutually exclusive: if event A occurs, then event B cannot occur
- Suppose we define the events:
 - A = no one has the characteristic (eg can run faster than 44 kph)
 - B = at least 1 person has the characteristicThen A and B are complementary events

$$\begin{aligned} &\Pr(\text{at least 1 person has the characteristic}) \\ &= 1 - \Pr(\text{no person has the characteristic}) \end{aligned}$$

$$\Pr(B) = 1 - \Pr(A)$$

Venn Diagrams

- Sample spaces and events are often presented in a visual display called a *Venn diagram*
- Use the following conventions
 - A *sample space* is represented by a *rectangle*
 - *Events* are represented by regions within the rectangle. This is usually done using *circles*



Conditional Probabilities

- The probability that event B will occur, given that an event A has occurred
- The notation for this conditional probability is $\Pr(B|A)$
- For any two events, A and B , the following relationship holds:

$$\Pr(B|A) = \frac{\Pr(A \text{ and } B)}{\Pr(A)}$$
- Versus independent events:

$$\Pr(A \text{ and } B) = \Pr(A) \times \Pr(B)$$
- *e.g: probability of passing Test 2 given a pass in Test 1?*
- Test 1 pass: 0.80 (of those sitting test)
- Test 2 pass: 0.63 (of those sitting test)
- Pass in Test 2 *AND* pass in Test 1: 0.51
- Pass Test 2 *GIVEN* pass in Test 1: 0.64
(If independent events: 0.50)

Conditional Probabilities

- The probability that event B will occur, given that an event A has occurred
- The notation for this conditional probability is $\Pr(B|A)$
- For any two events, A and B , the following relationship holds:

$$\Pr(B|A) = \frac{\Pr(A \text{ and } B)}{\Pr(A)}$$
- If two events A & B are *independent*:

$$\Pr(A \text{ and } B) = \Pr(A) \times \Pr(B)$$
- So:

$$\Pr(B|A) = \frac{\Pr(A \text{ and } B)}{\Pr(A)}$$
- That is, for ***independent events*** A and B the conditional probability that event B will occur, given that event A has occurred, is simply the probability that event B will occur

General Addition Law

- When two events are *not* mutually exclusive, use the following *general addition law*

$$\Pr(A \text{ or } B) = \Pr(A) + \Pr(B) - \Pr(A \text{ and } B)$$

- Remember: if the events A and B are **mutually exclusive**, $\Pr(A \text{ and } B) = 0$, hence:

$$\Pr(A \text{ or } B) = \Pr(A) + \Pr(B)$$

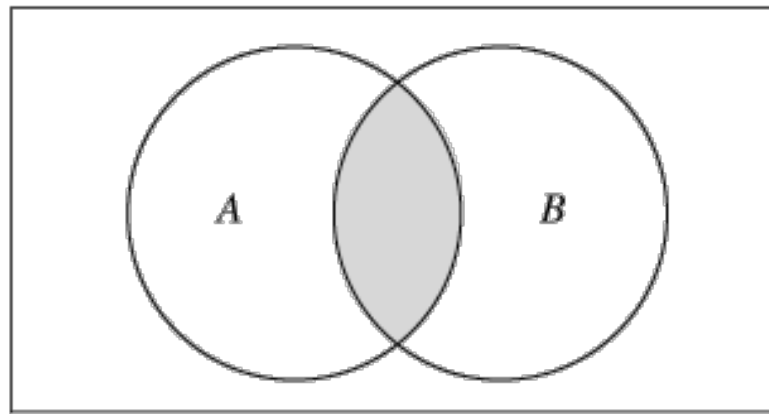
Venn Diagrams: Intersections

- The *intersection* of two events A and B is the set of all outcomes that are in *both* event A and event B . The notation is:

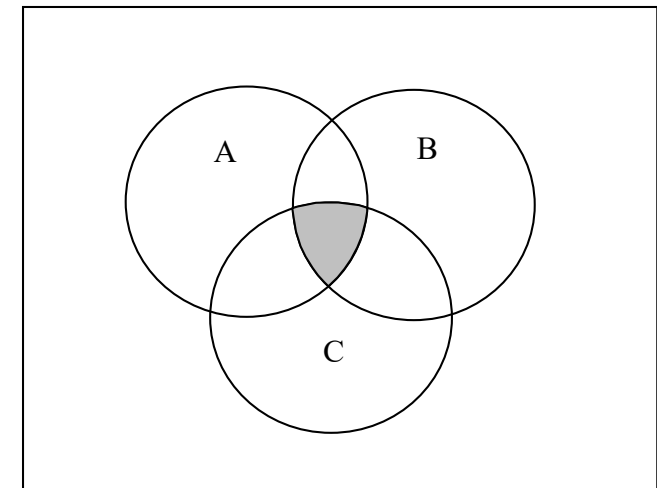
Intersection of event A and event $B = A \cap B$

Hence, we could write $\Pr(A \cap B)$ instead of $\Pr(A \text{ and } B)$

(d)



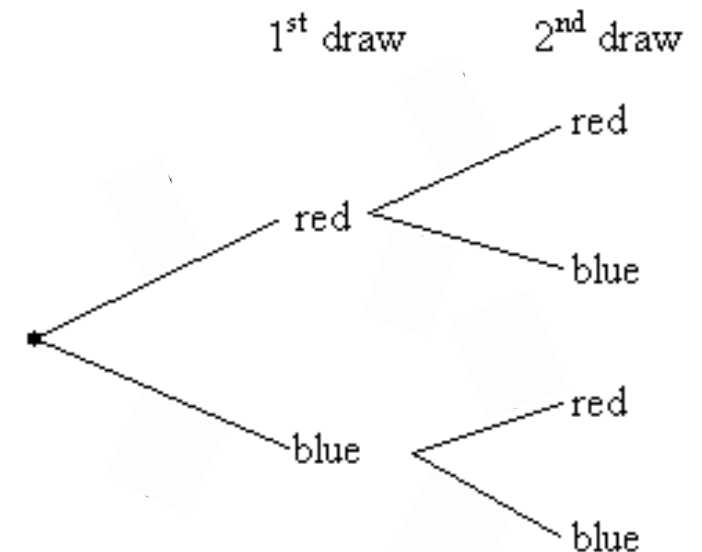
$A \cap B$



Intersection of events A and B and $C = A \cap B \cap C$

Probability Trees

- *Probability tree diagrams* can be useful for determining probabilities involving events that are *not* independent
- The joint probabilities for combinations of these events are found by multiplying the probabilities along the branches from the beginning of the tree
- If the events are not independent, the probabilities on the second tier of branches will be conditional probabilities, since their values will depend on what happened in the first event



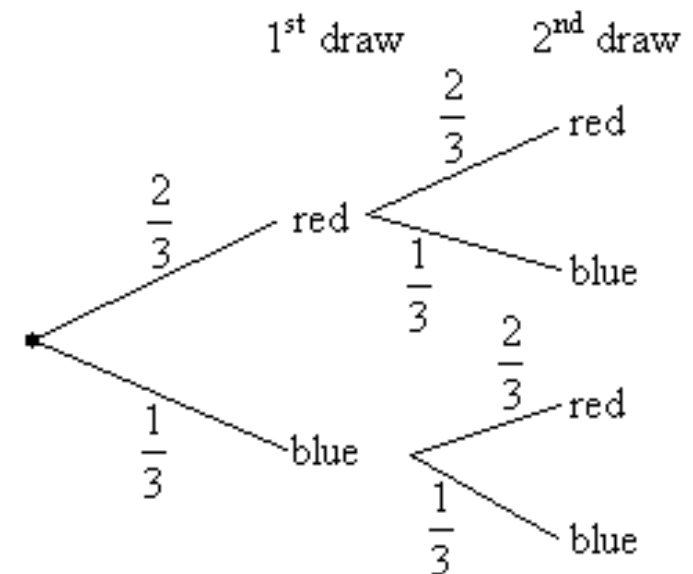
continued

Example: Drawing tokens

- *A box contains 4 red and 2 blue tokens. A token is drawn at random and then replaced. A second token is then drawn at random.*
- Calculate the probability of getting:

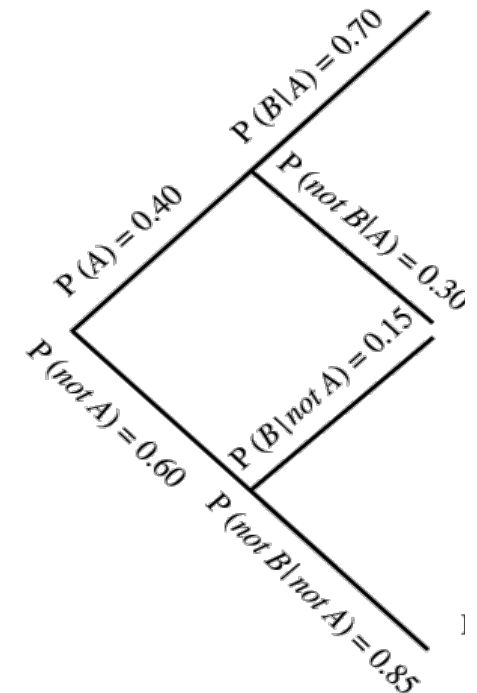
1. one red and one blue.
2. at least one blue.
3. two of the same color.

Sample space: {R,R; R,B; B,R; B,B}



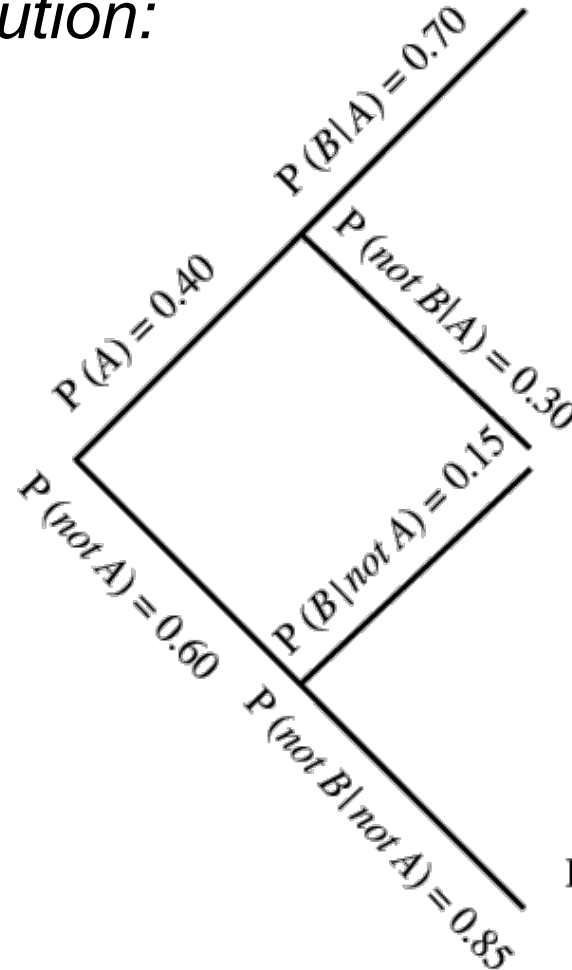
Probability Trees

- Example:
 - A clothing store has just imported a new range of suede jackets that it has advertised at a bargain price on a rack inside the store. The probability that a customer will try on a jacket is 0.40. If a customer tries on a jacket, the probability that he or she will buy it is 0.70. If a customer does *not* try on a jacket, the probability that he or she will buy it is 0.15
 - Calculate the probability that:
 - (a) a customer will try on a jacket and will buy it
 - (b) a customer will try on a jacket and will not buy it
 - (c) a customer will not try on a jacket and will buy it
 - (d) a customer will not try on a jacket and will not buy it



Probability Trees

Solution:



$$P(B|A) P(A) = 0.70 \times 0.40 = 0.28$$

$$P(\text{not } B|A) P(A) = 0.30 \times 0.40 = 0.12$$

$$P(B|\text{not } A) P(\text{not } A) = 0.15 \times 0.60 = 0.09$$

$$P(\text{not } B|\text{not } A) P(\text{not } A) = 0.85 \times 0.60 = \underline{0.51}$$

$$\text{Total} = 1.00$$

Summary

- We have looked at understanding elementary probability concepts
- We calculated the probability of events
- We distinguished between mutually exclusive, dependent and independent events
- We also looked at calculating conditional probabilities
- We understood and used the general addition law for probabilities
- We understood and applied Venn diagrams
- We understood and applied probability tree diagrams

Categorical Data

- Data are often *non-numerical*, in the sense that each individual observation is a description rather than a number
- Averages cannot be used in these circumstances
- Systems where the observations are descriptive (rather than numerical) are described as *categorical*, because the individuals are being classified into categories
- *E.g.*:
 - What is your gender?
 - What colour are your eyes?
 - Do you have a valid driver's licence?
 - Which suburb do you live in?
 - Do you have a fibre connection at home?
 - Who is your favourite lecturer?

Categorical Data

- Statistical questions can also involve categorical variables:
 - Are people who are avid followers of sport more likely to own a large-screen television than those who do not follow sport?
 - Does area of residence affect the likelihood of catching the bus?
 - Do people who live in particular part of a city have different radio preferences from those who live elsewhere?
 - Do males and females differ in their level of interest in watching the Rugby World Cup?

Categorical Data

- These questions may also be expressed as questions about differences between *proportions*, such as:
 - Does the proportion of individuals owning a large-screen television differ between avid followers of sport and others?
 - Does the proportion of people who take the bus to work differ from one area of residence to another?
 - Does the proportion of males interested in the Rugby World Cup differ from the proportion for females?
 - Does the proportion of wine-drinkers differ with age?

Single-variable categorical data

- It is common practice to have a standard form of presentation
- It is convenient to work with *frequency* data, that is data in which the *number* of occurrences of each category is recorded
- A frequency table is a table in which the number of occurrences of each category is recorded

Table 22.1 Outcomes of 60 rolls of a fair six-sided die							
Category	1	2	3	4	5	6	<i>Total</i>
Frequency	8	7	12	13	5	15	60
Class	—	—	—	—	—	—	—

(roll the dice 5 times each and record on the paper)

Contingency tables (“Cross-tab”)

- Some problems involve two categorical variables, and questions often arise about their relationship
- A *two-dimensional cross-tabulation* is where one variable is presented along the rows and the other variable down the columns

Table 22.3 A typical contingency table for area of residence and fibre penetration					
Fibre \ Live	North	South	East	West	<i>Total</i>
Yes	52	47	105	34	238
No	28	63	35	36	162
<i>Total</i>	80	110	140	70	400

Contingency tables

- Contingency tables have characteristics that are common to all such tables. These include:
 - The final column is a total column
 - The final row is a total row
 - The final (bottom-right) cell is the Grand Total
 - It generally does not matter which variable is along the columns and which is along the rows
 - Frequencies must add up within each row
 - Frequencies must add up within each column
 - Can also be formatted as percentages (of row / column / total)

Hypothesis Testing: Proportions

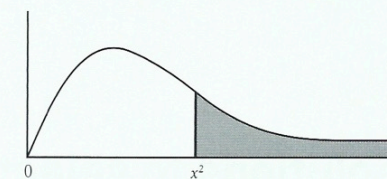
- An **hypothesis** is an unproven proposition or supposition that tentatively explains certain facts or phenomena.
 - Empirically testable
- A **null hypothesis** is a statement about a status quo.
 - Any change from what has been thought to be true will be due entirely to random sampling error.
- An **alternative hypothesis** is a statement indicating the opposite of the null hypothesis.

Analysis of single-variable problems

- Is an *observed* set of categorical data reasonably consistent with what was *expected* (eg by theory, best practice, calibrated machine)
- The steps involved are known as a *goodness-of-fit* test, using *hypotheses*
- 1. Construct a *null hypothesis* and *alternate hypothesis* for the problem. This usually takes the general form of:
 - H_0 : There is no difference between the observed frequencies and the expected frequencies
 - H_1 : There IS a difference between the observed frequencies and the expected frequenciesE.g: H_0 : The proportions of men and women in the Glenfield area are equal
 - H_1 : The proportions of men and women are not equal

Hypothesis Testing: Proportions

- **Significance level** is the critical probability in choosing between the null and alternative hypotheses
 - Eg alpha = 0.05: happy to accept the probability of being wrong 1 time in 20
 - For proportions, we use the χ^2 distribution ('chi-square', where 'chi' = 'ky' [rhymes with 'pie'])



This table gives the critical values for various degrees of freedom ν and significance levels α

Table 9 Critical values for the chi-square distribution

α ν	0.10	0.05	0.025	0.01	0.005	0.001	α ν
1	2.706	3.841	5.024	6.635	7.879	10.827	1
2	4.605	5.991	7.378	9.210	10.597	13.815	2
3	6.151	7.815	9.348	11.345	12.838	16.266	3
4	7.779	9.488	11.143	13.277	14.860	18.466	4
5	9.236	11.070	12.832	15.086	16.250	20.515	5

Chi-square: single-variable

2. Obtain the *observed frequencies* from the data of the problem
3. Determine the *expected frequencies*; these are ones we might ‘expect’ to occur if H_0 were true
4. Calculate the measure of the discrepancy between the observed and expected frequencies by using the *chi-square test* statistic

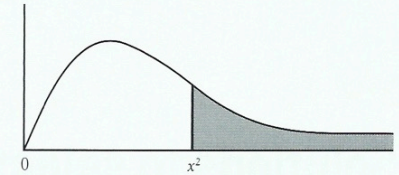
$$\chi^2 = \sum_{\text{all categories}} \frac{(\text{observed frequency} - \text{expected frequency})^2}{\text{expected frequency}}$$

- Since the square of a number can never be negative, the value of a χ^2 -test statistic can also never be negative

5. Associated with the test statistic are *degrees of freedom*. Determine the degrees of freedom for a goodness-of-fit test using:

$$\text{Degrees of freedom} = \text{number of categories} - 1$$

Chi-square: single-variable (p. 787)



This table gives the critical values for various degrees of freedom ν and significance levels α

Table 9 Critical values for the chi-square distribution

$\alpha \backslash \nu$	0.10	0.05	0.025	0.01	0.005	0.001	$\alpha \backslash \nu$
1	2.706	3.841	5.024	6.635	7.879	10.827	1
2	4.605	5.991	7.378	9.210	10.597	13.815	2
3	6.151	7.815	9.348	11.345	12.838	16.266	3
4	7.779	9.488	11.143	13.277	14.860	18.466	4
5	9.236	11.070	12.832	15.086	16.750	20.515	5
6	10.645	12.592	14.449	16.812	18.548	22.457	6
7	12.017	14.067	16.013	18.475	20.278	24.321	7
8	13.362	15.507	17.535	20.090	21.955	26.124	8
9	14.684	16.919	19.023	21.666	23.589	27.877	9
10	15.987	18.307	20.483	23.209	25.188	29.588	10
11	17.275	19.675	21.920	24.725	26.757	31.264	11
12	18.549	21.026	23.337	26.217	28.300	32.909	12
13	19.812	22.362	24.736	27.688	29.819	34.527	13
14	21.064	23.685	26.119	29.141	31.319	36.124	14
15	22.307	24.996	27.488	30.578	32.801	37.698	15
16	23.542	26.296	28.845	32.000	34.267	39.252	16
17	24.769	27.587	30.191	33.409	35.718	40.791	17
18	25.989	28.869	31.526	34.805	37.156	42.312	18
19	27.204	30.144	32.852	36.191	38.582	43.819	19
20	28.412	31.410	34.170	37.566	39.997	45.314	20
21	29.615	32.671	35.479	38.932	41.401	46.796	21
22	30.813	33.924	36.781	40.289	42.796	48.268	22
23	32.007	35.172	38.076	41.638	44.181	49.728	23
24	33.196	36.415	39.364	42.980	45.558	51.179	24
25	34.382	37.652	40.646	44.314	46.928	52.619	25
26	35.563	38.885	41.923	45.642	48.290	54.051	26
27	36.741	40.113	43.195	46.963	49.645	55.475	27
28	37.916	41.337	44.461	48.278	50.994	56.892	28
29	39.087	42.557	45.722	49.588	52.335	58.301	29
30	40.256	43.773	46.979	50.892	53.672	59.702	30
40	51.805	55.758	59.342	63.691	66.766	73.403	40
50	63.167	67.505	71.420	76.154	79.490	86.660	50
60	74.397	79.082	83.298	88.379	91.952	99.608	60
70	85.527	90.531	95.023	100.425	104.215	112.317	70
80	96.578	101.979	106.629	112.329	116.321	124.839	80
90	107.565	113.145	118.136	124.116	128.299	137.208	90
100	118.498	124.342	129.561	135.807	140.170	149.449	100

6. Obtain the *critical value*, from Chi-square distribution (Table 9, p.787).

Two pieces of information are required: the *degrees of freedom* (down the left-hand column) and the *significance level* (across the top row)

☞ 0.05 is a commonly-used significance level (α [alpha])

7. Compare the calculated value of χ^2 with the critical value from Table 9 (χ^2_c)

If $\chi^2 < \chi^2_c$, we cannot reject H_0

If $\chi^2 > \chi^2_c$, we reject H_0

8. Draw an appropriate conclusion

Chi-square: single-variable

Example:

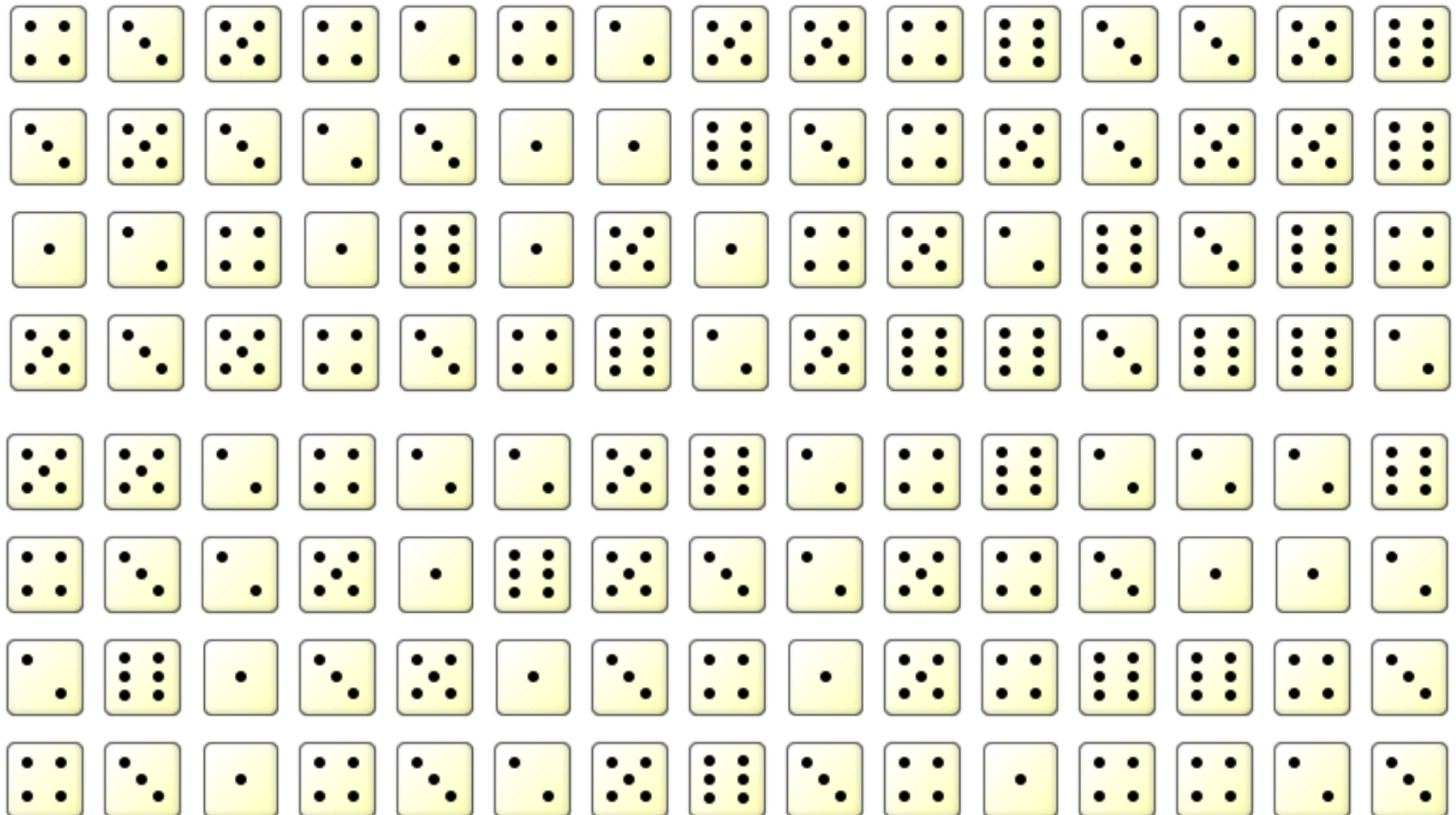
Suppose that a data analyst is presented with a six-sided die and asked to determine whether it is ‘fair’ – that is, whether it is equally likely that the outcome will be a 1, 2, 3, 4, 5 or 6 when the die is tossed. The die is rolled a total of 60 times. The outcomes are shown in the following table:

Die	Observed (O)	Expected (E)
1	8	10
2	7	10
3	12	10
4	13	10
5	5	10
6	15	10
Total	60	60

If the die is really fair, there is a $1/6$ probability that any given face will appear at any roll. Thus, in a loose sense, the 60 rolls would be ‘expected’ to yield $60 \times 1/6 = 10$ occurrences of each face

Dice rolls: Random.Org/dice

120 rolls



Dice rolls: Random.Org/dice

120 rolls

	Die face:					
Row						
1	0	2	3	4	4	2
2	2	1	5	1	4	2
3	4	2	1	3	2	3
4	0	2	3	2	3	5
5	0	7	0	2	3	3
6	3	3	3	2	3	1
7	3	1	3	3	2	3
8	2	2	4	5	1	1
Observed	14	20	22	22	22	20

Chi-square: single-variable

Solution:

Step 1: H_0 : The die *is* fair;
 H_1 : The die *is not* fair

Step 2: The observed frequencies are the actual values obtained for each category; that is 14, 20, 22, 22, 22, 20

Step 3: Since H_0 assumes that the die is fair, the expected frequency for each category is the same, that is, $120 \times 1/6 = 20$

Dice rolls: Random.Org/dice

120 rolls

	Die face:					
Row						
1	0	2	3	4	4	2
2	2	1	5	1	4	2
3	4	2	1	3	2	3
4	0	2	3	2	3	5
5	0	7	0	2	3	3
6	3	3	3	2	3	
7	3	1	3	3	2	3
8	2	2	4	3	1	1
Observed	14	20	22	22	22	20
Expected	20	20	20	20	20	20
O-E	-6	0	2	2	2	0

Calculating residuals from the expected value of 20 (since we have 120 rolls ÷ 6 dice faces)

$$\chi^2 = \sum_{\text{all categories}} \frac{(\text{observed frequency} - \text{expected frequency})^2}{\text{expected frequency}}$$

Chi-square: single-variable

$$\chi^2 = \sum_{\text{all categories}} \frac{(\text{observed frequency} - \text{expected frequency})^2}{\text{expected frequency}}$$

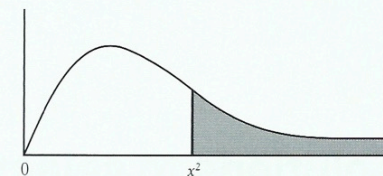
Step 4: For the die, the calculations required for the χ^2 -test statistic are:

$$\begin{aligned} \chi^2 &= \frac{(14-20)^2}{20} + \frac{(20-20)^2}{20} + \frac{(22-20)^2}{20} + \\ &\quad \frac{(22-20)^2}{20} + \frac{(22-20)^2}{20} + \frac{(20-20)^2}{20} \\ &= 0.4 + 0 + 0.2 + 0.2 + 0.2 + 0 \\ &= 2.4 \end{aligned}$$

Die	Observed (O)	Expected (E)	Observed - Expected (O - E)	Squared Differences (O - E) ²	$\frac{(O - E)^2}{E}$
1	14	20	-6	36	1.8
2	20	20	0	0	0
3	22	20	2	4	0.2
4	22	20	2	4	0.2
5	22	20	2	4	0.2
6	20	20	0	0	0
Total	120	120	0	48	2.4

Step 5: Since there are 6 die categories, the degrees of freedom (*df*) are $6 - 1 = 5$

Chi-square: single-variable



This table gives the critical values for various degrees of freedom ν and significance levels α

Step 6: If significance level $\alpha = 0.05$ is desired, we go to the degrees of freedom row 5 and column 0.05 to obtain a critical value of $\chi^2_c = 11.07$

Step 7: For the die, we have:

$$\chi^2 = 7.6 \text{ and } 7.6 < 11.07$$

Therefore we cannot reject H_0

Step 8: Since we cannot reject H_0 , the conclusion is that it is probable that the die is fair. That is, the evidence of the outcomes of the rolls does not give us grounds to conclude that the die is not fair

Table 9 Critical values for the chi-square distribution

$\alpha \backslash \nu$	0.10	0.05	0.025	0.01	0.005	0.001	$\alpha \backslash \nu$
1	2.706	3.841	5.024	6.635	7.879	10.827	1
2	4.605	5.991	7.378	9.210	10.597	13.815	2
3	6.151	7.815	9.348	11.345	12.838	16.266	3
4	7.779	9.488	11.143	13.277	14.860	18.466	4
5	9.236	11.070	12.832	15.086	16.750	20.515	5
6	10.645	12.592	14.449	16.812	18.548	22.457	6
7	12.017	14.067	16.013	18.475	20.278	24.321	7
8	13.362	15.507	17.535	20.090	21.955	26.124	8
9	14.684	16.919	19.023	21.666	23.589	27.877	9
10	15.987	18.307	20.483	23.209	25.188	29.588	10
11	17.275	19.675	21.920	24.725	26.757	31.264	11
12	18.549	21.026	23.337	26.217	28.300	32.909	12
13	19.812	22.362	24.736	27.688	29.819	34.527	13
14	21.064	23.685	26.119	29.141	31.319	36.124	14
15	22.307	24.996	27.488	30.578	32.801	37.698	15
16	23.542	26.296	28.845	32.000	34.267	39.252	16
17	24.769	27.587	30.191	33.409	35.718	40.791	17
18	25.989	28.869	31.526	34.805	37.156	42.312	18
19	27.204	30.144	32.852	36.191	38.582	43.819	19
20	28.412	31.410	34.170	37.566	39.997	45.314	20
21	29.615	32.671	35.479	38.932	41.401	46.796	21
22	30.813	33.924	36.781	40.289	42.796	48.268	22
23	32.007	35.172	38.076	41.638	44.181	49.728	23
24	33.196	36.415	39.364	42.980	45.558	51.179	24
25	34.382	37.652	40.646	44.314	46.928	52.619	25
26	35.563	38.885	41.923	45.642	48.290	54.051	26
27	36.741	40.113	43.195	46.963	49.645	55.475	27
28	37.916	41.337	44.461	48.278	50.994	56.892	28
29	39.087	42.557	45.722	49.588	52.335	58.301	29
30	40.256	43.773	46.979	50.892	53.672	59.702	30
40	51.805	55.758	59.342	63.691	66.766	73.403	40
50	63.167	67.505	71.420	76.154	79.490	86.660	50
60	74.397	79.082	83.298	88.379	91.952	99.608	60
70	85.527	90.531	95.023	100.425	104.215	112.317	70
80	96.578	101.979	106.629	112.329	116.321	124.839	80
90	107.565	113.145	118.136	124.116	128.299	137.208	90
100	118.498	124.342	129.561	135.807	140.170	149.449	100

Dice rolls: Random.Org/dice

120 rolls

	Die face:					
Row						
1	0	2	3	4	4	2
2	2	1	5	1	4	2
3	4	2	1	3	2	3
4	0	2	3	2	3	5
5	0	7	0	2	3	3
6	3	3	3	2	3	1
7	3	1	3	3	2	3
8	2	2	4	5	1	1

Observed	14	20	22	22	22	20
Expected	20	20	20	20	20	20
O-E	-6	0	2	2	2	0
(O-E)²	36	0	4	4	4	0
/E	1.8	0	0.2	0.2	0.2	0
χ^2	2.4					
d.f	5					
χ^2_c	11.08					

Chi-square: cross-tab analysis

- The χ^2 technique can be generalised to the case where two variables are involved
 - The data will be in the form of a contingency table with any number of rows and columns
1. Construct the null hypothesis for the problem. This usually takes the general form that the two variables are *independent* or that there is *no relationship* between them

H_0 : The two variables *are* independent, or

H_0 : There is *no* relationship between the two variables

The alternative hypothesis (using a two-sided alternative) would be:

H_1 : The two variables are *not* independent, or

H_1 : There *is* a relationship between the two variables

Chi-square: cross-tab analysis

2. Identify the *observed frequencies* from the data of the problem. There will be one observed frequency for each cell of the contingency table
3. Calculate the *expected frequencies*, those that we might 'expect' to occur if H_0 were true. For each cell of the contingency table the expected frequency can be calculated as:

$$= \frac{\text{total frequency for that column} \times \text{total frequency for that row}}{\text{grand total frequency}}$$

The *grand total frequency* can be found in the bottom right-hand corner of the table

This assumes independence between the two variables

$$\Pr(A \text{ and } B) = \Pr(A) \times \Pr(B)$$

Chi-square: cross-tab analysis

4. Calculate the measure of the discrepancy between the observed and expected frequencies using the χ^2 *test statistic*. The formula is:

$$\chi^2 = \sum_{\text{all cells}} \frac{(\text{observed frequency} - \text{expected frequency})^2}{\text{expected frequency}}$$

Note that a calculation is required for each cell of the table.

5. Determine the degrees of freedom for the contingency table

$$\text{Degrees of freedom} = (\text{number of rows} - 1) \times (\text{number of columns} - 1)$$

Chi-square: cross-tab analysis

6. Obtain the critical value from Table 9 (p787), using both the degrees of freedom and the desired significance level
7. Compare the value of χ^2 that you calculated with the critical value from Table 9
 - If $\chi^2 <$ the critical value, we *cannot reject* H_0
 - If $\chi^2 >$ the critical value, we *can reject* H_0
8. Based on the outcome of Step 7, draw an appropriate conclusion

Lab 6

115.109: Topic 6 Lab Worksheet: Sustainability data PivotTables

Due date: 31 August 2026

Vers

Task 1: Create a cross-tabulation via PivotTable of Gender by Age

In the SustainabilityData sheet, select Insert menu > click on PivotTa
Drag the Age variable into the Rows box, and Gender into the Colum
The resulting table will look ugly. Click on the Column filter button
Paste special > values your result here:

1a: Cross-tab of Gender and Age

	1 (Male)	2 (Female)	Male + Female	Total
1 (18-24)			0	#
2 (25-34)			0	#
3 (35-44)			0	#
4 (45-54)			0	#
5 (55-64)			0	#
6 (65+)			0	#
Total	0	0	0	#

Rationale:

This lab again uses the SustainabilityData dataset from Perceptive provided for the Report.

The first task is to explore what our sample looks like. We explored the Gender, Age and Income breakdown in Lab 5, but this time we will look for any relationships between these variables, and to see if there is any bias in sample selection.

We contrast percentage of row and percentage of column, calculated manually, as different views of the same data.

In Task 1b we look at a crosstab of Age and Income, and then in 1c reformat the PivotTable as % within column. In Task 1d we use the Group function to reduce down the Age and Income categories (also known as "on-the-fly" recoding, since these groupings are not permanently stored in the dataset.)

In Tasks 2 and 3 we calculate the Chi-Square statistic for some of the tables in Task 1, to determine if there is any sample bias.

Finally in Task 4 we investigate the data further, by compiling descriptive statistics (means) for some of our independent variables (IVs): "HowOftenBuy"; "PayMore". "ReturnPackaging" and "BottleDepositScheme". We then contrast subsamples, by looking at whether the means differ for respondents by gender and then income. (You can optionally also explore age, and you can also combine Income & Gender or Income & Age into a single multiway table if you like: means by Income AND Gender.

Lab 6

115.109: Topic 6 Lab Worksheet: Sustainability data PivotTables

Due date: 31 August 2026

Version: 1 (6 August 2026)

3

☐ 1: Male 2: Female 3: Gender c
☐ Which gender do you identify

Task 1: Create a cross-tabulation via PivotTable of Gender by Age, as follows:

In the SustainabilityData sheet, select Insert menu > click on PivotTable button, for the range \$B\$4:\$CM\$1046

Drag the Age variable into the Rows box, and Gender into the Columns, and either of them into the Values box.

The resulting table will look ugly. Click on the Column filter button **Column Labels** to remove the last three selections

Paste special > values your result here:

1a: Cross-tab of Gender and Age

				% within row			% within column	
	1 (Male)	2 (Female)	Male + Female	Age % Male	Age % Female	Age % Total	Male % Age	Female % Age
1 (18-24)			0	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
2 (25-34)			0	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
3 (35-44)			0	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
4 (45-54)			0	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
5 (55-64)			0	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
6 (65+)			0	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
Total	0	0	0	#DIV/0!	#DIV/0!		#DIV/0!	#DIV/0!

PivotTable F

FIELD NAME

☐ Variable
☒ Gender
☒ Age
☐ Region

Filters



Lab 6

Count of Age	Column Labels		
Row Labels	1	2	Grand Total
1	13	54	67
2	94	172	266
3	91	129	220
4	58	84	142
5	64	71	135
6	128	81	209
Grand Total	448	591	1039

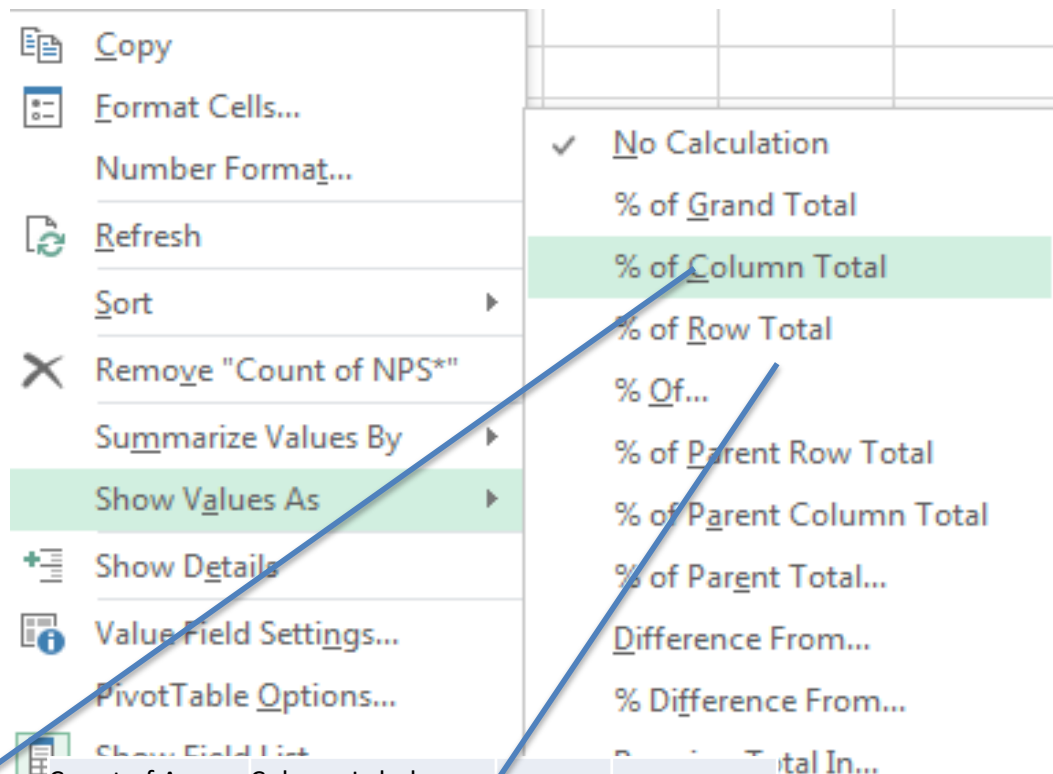
Age
1=18-24
2=25-34
3=35-44
4=45-54
5=55-64
6=65+

Gender
1=Male
2=Female
(3=Diverse)

Count of Age	Column Labels		
Row Labels	1	2	Grand Total
1		9.14%	6.45%
2	20.98%	29.10%	25.60%
3	20.31%		21.17%
4	12.95%	14.21%	13.67%
5	14.29%	12.01%	12.99%
6	28.57%	13.71%	20.12%
Grand Total	100.00%	100.00%	100.00%

PivotTable Fields

Choose fields to add to report:



Count of Age	Column Labels		
Row Labels	1	2	Grand Total
1		80.60%	100.00%
2	35.34%	64.66%	100.00%
3	41.36%		100.00%
4	40.85%	59.15%	100.00%
5	47.41%	52.59%	100.00%
6	61.24%	38.76%	100.00%
Grand Total	43.12%	56.88%	100.00%



Lab 6

1b: Cross-tab of Age and Income									
Now repeat the process using a cross-tab of Age & Income.									
Copy the PivotTable in the new sheet, paste anywhere beside or beneath, and modify. Paste Special your result here:									
	1: Up to \$30,000	2: \$30,001 - \$50,000	3: \$50,001 - \$70,000	4: \$70,001 - \$100,000	5: \$100,001- \$130,000	6: Over \$130,000	7: Rather not say	Grand Total (All Income)	Age % All Income
1 (18-24)								0	#DIV/0!
2 (25-34)								0	#DIV/0!
3 (35-44)								0	#DIV/0!
4 (45-54)								0	#DIV/0!
5 (55-64)								0	#DIV/0!
6 (65+)								0	#DIV/0!
Total	0	0	0	0	0	0	0	0	

Hint: it doesn't matter if 'Count of Age' - it shows drag to the Values box. But if the 'Values' says a to change what is being choose Field Settings >

1c: Repeat cross-tab of Age and Income, but this time use '% of Column Total' as the Value Field > Show data as... option									
Copy the table from 1b (beside or beneath in your new sheet), and modify. Paste special > Values the result here:									
	1: Up to \$30,000	2: \$30,001 - \$50,000	3: \$50,001 - \$70,000	4: \$70,001 - \$100,000	5: \$100,001- \$130,000	6: Over \$130,000	7: Rather not say	Grand Total (All Income)	
1 (18-24)									
2 (25-34)									
3 (35-44)									
4 (45-54)									
5 (55-64)									
6 (65+)									
Total	0%	0%	0%	0%	0%	0%	0%	0%	

Hint: Select Field Settings > Show data as ... % Column.

 Check for yourself what the difference is if you choose % Row Total instead.

 If the rows add up to 100%, then you can compare within the column.

 If the columns add up to 100%, then you can compare within the row.

PivotTable: Grouping

1d: Repeat cross-tab of Age and Income, but this time combine both Age and Income categories into fewer classes (use Counts, not % of columns)

Scroll right for hints! ==>

Copy the table from 1c (beneath in your new sheet), and modify. Paste special the result here:

	Lower (<\$50,000)	Middle (\$50-100k)	(>\$100k) - \$70,000	Not said	Size by Age (All Income)
Younger (18-34)					
Middle (35-54)					
Older (55+)					
Subsample size by Income	0	0	0	0	0

Hint: Remove the Age/Gender table, and modify the Age/Income PivotTable (while you can group Income, Excel will complain when you try to group by Age if you have two tables with Age in it). Select the rows you wish to group, then right-click any selected row and choose Group.... (or use the Group Selection button in the ribbon)

1) Since there are six Age categories, let us group them into three age categories: 18-24 + 25-34 into "Younger"; 35-44 + 45-54 into "Middle"; and 55-64 and 65+ into "Older". This will create a new variable called Age2, located at the bottom of the variable list of the cached dataset for the PivotTable (not in the actual source table). Repeat for the other ages into Group 2 (Ages 3 & 4) and Group 3 (Ages 5 & 6), and rename the resulting Groups into Younger, Middle and Older as above. Once you have Age2, you can remove Age from the PivotTable.

2) To group (Household)Income, in the most recent version of Excel you can similarly group within columns (in older versions you can only group in rows, and so would need to swap rows and columns around: drag Age2 to the Columns box, and Income to the Rows). Repeat as above, but for:

"Lower" (income groups 1 & 2, <\$50k); "Middle" (3&4, \$50-100k) and "Higher" (5&6, >\$100k) in new variable Income2. Remove Income. Make sure Age2 is in rows, and Income2 is in columns (to look like the table to the left). Copy and Paste Special > Values the results to this table, rather than pasting the PivotTable.

35 If you have problems grouping this way, you can use an alternative method: clicking on just the first row or column header (category 1) and right-click > Group...

36 This allows you to select how the grouping works automatically: group by 2 (eg 1&2, 3&4), leaving the last group being just 7 (Excel incorrectly calls this >7 but it should be ≥7)

37 Repeat for the columns, and then filter out the age category 7. **Please note:** using this method, you do NOT get new variables Age2 and HouseholdIncome2. No idea why the difference!

38

39 Count of Household Income

40 Row Labels

41 1 2 3 4 5 6 7 Grand Total

42 10 9 15 10 7 2 15 68

43 2 12 30 52 66 42 27 37 266

44 3 4 26 37 45 36 26 36 220

45 4 8 22 22 23 15 21 21 142

46 5 0 18 15 17 16 15 25 136

47 6 1 62 29 20 16 8 43 209

48 7 1 1 1 1 1 1 1 7

49 Grand Total 6 167 170 181 132 99 177 1042

50

51

52

Count of Household Income

Row Labels

1 2 3 4 5 6 7 Grand Total

1-2 22 39 67 76 49 29 52 334

3-4 ? X 51 47 57 362

5-6 32 23 68 345

>7 1 1

Grand Total 132 99 177 1042

Count of Age

Row Labels

1-2 3-4 5-6 >7 Grand Total

>7 1 1 1 1 4

1-2 61 143 78 52 334

3-4 80 127 98 57 362

5-6 141 81 55 68 345

Grand Total 283 351 231 177 1042

Count of Age

Row Labels

1-2 3-4 5-6 >7 Grand Total

1-2 61 143 78 52 334

3-4 80 127 98 57 362

5-6 141 81 55 68 345

Grand Total 282 351 231 177 1041

Lab 6

Task 2: Calculate expected values for the table in Task 1 (reproduced below); then the residuals; then the squared residuals; and finally Chi-Square

Observed				Row	Residuals: Observed - Expected			
Observed	Column C	D	E		Expected	Column Labels		
Row Labels			Grand Total		Row Labels	Male	Female	Grand Total
1	13	54	67	58	18-24	-15.9	15.9	0.0
2	94	172	266	59	25-34	-20.7	20.7	0.0
3	91	129	220	60	35-44	-3.9	3.9	0.0
4	58	84	142	61	45-54	-3.2	3.2	0.0
5	64	71	135	62	55-64	5.8	-5.8	0.0
6	128	81	209	63	65+			0.0
Grand Total	448	591	1039	64	Grand Total	0.0	0.0	0.0

$$= C\$64 * \$E58 / \$E\$64$$

Expected: row totals x column totals / totals

Row Labels	Male	Female	Grand Total
18-24	28.9	38.1	67.0
25-34	114.7	151.3	266.0
35-44	94.9	125.1	220.0
45-54	61.2		142.0
55-64	58.2	76.8	135.0
65+	90.1	118.9	209.0
Grand Total	448.0	591.0	1039.0

$$\chi^2 = \sum_{\text{all categories}} \frac{(\text{observed frequency} - \text{expected frequency})^2}{\text{expected frequency}}$$

Squared residuals / Expected

Count of Age	Male	Female	Chi Square
18-24	8.74		
25-34	3.73	2.83	
35-44	0.16	0.12	
45-54	0.17	0.13	
55-64	0.58	0.44	
65+	15.92	12.07	
Grand Total			

Lab 6

Task 2: Calculate expected values for the table in Task 1 (reproduced below); then the residuals; then the squared residuals; and finally Chi-Square

Observed	Column C	D	Row
Column Labels			
Row Labels	1	2	Grand Total
1	13	54	67
2	94	172	266
3	91	129	220
4	58	84	142
5	64	71	135
6	128	81	209
Grand Total	448	591	1039

Expected	Column Labels		
Row Labels	Male	Female	Grand Total
18-24	-15.9	15.9	0.0
25-34	-20.7	20.7	0.0
35-44	-3.9	3.9	0.0
45-54	-3.2	3.2	0.0
55-64	5.8	-5.8	0.0
65+			0.0
Grand Total	0.0	0.0	0.0

$$\chi^2 = \sum_{\text{all categories}} \frac{(\text{observed frequency} - \text{expected frequency})^2}{\text{expected frequency}}$$

Expected: row totals x column totals / totals

Expected	Column Labels		
Row Labels	Male	Female	Grand Total
18-24	28.9	38.1	67.0
25-34	114.7	151.3	266.0
35-44	94.9	125.1	220.0
45-54	61.2		142.0
55-64	58.2	76.8	135.0
65+	90.1	118.9	209.0
Grand Total	448.0	591.0	1039.0

Squared residuals / Expected

Count of Age	Column Labels		
Row Labels	Male	Female	Chi Square
18-24	8.74		
25-34	3.73	2.83	
35-44	0.16	0.12	
45-54	0.17	0.13	
55-64	0		
65+	15		
Grand Total			

Degrees of freedom: (# columns - 1) * (# rows - 1) = (2-1)(6-1) = 5

Critical value of chi-square with 5 d.f: 11.07 {=CHIINV(0.05,5)}

Conclusion: chi-square is significant (Chi-square > critical value of chi-square)
(variables *are* related: 'Age' and 'Gender' are *not* independent of each other)

Table 9 Critical values for the

α	0.10	0.05
1	2.706	3.841
2	4.605	5.991
3	6.151	7.815
4	7.779	9.488
5	9.236	11.070
6	10.645	12.592

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α	β
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The graph shows a function $f(x)$ on the interval $[0, a]$. The area under the curve from $x=a$ to $x=b$ is shaded.

3e: Squared Residuals over Expected (O-E)*(O-E) / E			
	Lower (<\$50,000)	Middle (\$50-100k)	Upper (>\$100k) - \$70,000
Younger (18-34)			
Middle (35-54)			
Older (55+)			
			Chi-Square
			0.00

Lab 6

Task 4: Calculate the mean rating for columns AY, AZ, BA, BB, BC and BD in the datasheet:

First for all respondents, and then for Male vs Female, then Lower vs Middle vs Higher income groups:

		All respondents	Male	Female	Lower (<\$50k)	Middle (\$50-100k)	Higher (>\$100k)
	Variable	Mean	Mean	Mean	Mean	Mean	Mean
AY	HowOftenBuy						
AZ	PayMore						
BA	WillingPayMore						
BB	AcceptablePayMore						
BC	ReturnPackaging						
BD	BottleDeposit						
	(Sub)Sample size						
	Interpret the table:						
	<<Table Interpretation>>						

How: Create a new PivotTable (or heavily modify another copy)

1) Select the six variables (HowOftenBuy to BottleDeposit), putting each in the Σ Values box. Change the Σ Values from **Count** to **Average** by clicking on the Σ icon (Mac) or arrow (PC).

Format so that the means only have one decimal place, and copy into C91:C94.

2) Copy your whole PivotTable and paste beneath. Add Gender to the Columns box, and filter out everything except 1 (Male) and 2 (Female). To get the subsample size, add Gender to the Σ Values box, leaving as Count of Gender.

3) Copy the second PivotTable and paste beneath (or recreate a new one). This time add Income2 to the Columns box, and filter out everything except 1 (Lower) and 2 (Middle) and 3 (Higher). To get the subsample size, add Income2 to the Σ Values box, leaving as Count of Income2. Also add the variable Income2 into the Σ Values box to get the subsample count.

BONUS TASK 5

For the Income2 PivotTable: add Gender back into Columns, this time moving Income2 back into Rows. This creates a combination table, where you can contrast the means for the six variables by Income AND Gender.
Paste your combination table below.

<<End of Lab 6>>

AI Declaration for Lab 6: complete the declaration when submitting the file

Summary

- We have understood
 - the meaning of a categorical variable
 - the difference between a single-variable problem and a two-variable problem
- We constructed
 - a table for a single-variable problem
 - a contingency table for a two-variable problem
- We analysed single-variable data
- We analysed two-variable data